情報論

～英文に翻訳～

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Abstract の翻訳

Researchers have actively studied Multi-Agent Simulation (MAS) across various application domains as a powerful technique to construct virtual social environments, reproduce complex social phenomena, and analyze the behavior of social systems. Recently, MAS has helped evaluate various policies before and after implementation during the COVID-19 pandemic; for instance, Kurahashi [23] used it to predict infection spread based on vaccination rates. Furthermore, Sánchez [22] employed MAS to analyze how urban bicycle-sharing services impact human mobility and to pre-validate effective operational methods for practical implementation. As these examples illustrate, MAS shows increasing promise as a key technology to support planning and implementing policies that broadly appeal to society and systems embedded within it.

MAS models entities like humans or intelligent robots as agents that act and make decisions, and it simulates chains of their interactions. Consequently, how we construct these agent models becomes a critical technical challenge that determines the simulation's performance. When MAS aims to understand complex social phenomena or analyze social systems, an approach that constructs coarse-grained, more abstract agent models, such as one following the KISS principle [Axelrod 84], proves effective. This is because such models make it easier to analyze the causal relationships between individual behaviors, their interactions, and the emergent collective phenomena.

However, when MAS aims to design or verify policies and systems for real-world application, we need fine-grained, more concrete agent models to provide information useful for administrative decision-making and for gaining public acceptance. This necessity arises because the behavior of abstract models, which often disregard the specific characteristics of individual actors, can diverge from stakeholder perspectives. This divergence makes it difficult to meticulously verify the societal impact of novel policies and systems.

はじめにの翻訳

Multi-Agent Simulation (MAS) is a powerful technique for building virtual social environments to reproduce complex social phenomena and analyze the behavior of social systems. It has been applied in various research fields. Recently, MAS has been used to evaluate countermeasures during the COVID-19 pandemic, such as predicting the spread of infection based on vaccination rates [Kurahashi 23]. It has also been used to analyze the impact of urban bicycle-sharing services on human mobility and to validate effective operational strategies [Sanchez 22]. These examples highlight MAS as an emerging and promising technology to support the design and implementation of policies and systems with broad societal impact.

In MAS, entities that act and make decisions, such as humans or intelligent robots, are modeled as agents, and the simulation computes the chain of their interactions. Consequently, the method for building these agent models is a critical technical challenge that determines the simulation's performance. When the goal of MAS is to understand complex social phenomena or analyze social systems, a coarse-grained, more abstract approach to agent modeling, such as following the KISS principle [Axelrod 84], is effective. This approach simplifies the analysis of causal relationships between individual behaviors, their interactions, and the resulting emergent collective phenomena. However, when the goal is to design and validate policies or systems for real-world application, fine-grained, more concrete agent models are necessary to provide information that can inform administrative decisions and gain public acceptance. The behavior of abstract models, which ignore the specific characteristics of individual actors, can diverge from stakeholder perspectives, making it difficult to thoroughly examine the societal impact of novel policies and systems.

A key challenge in building fine-grained agent models lies in the methodology and development cost of creating models that can reproduce diverse and detailed behaviors for a wide range of simulation targets. It is difficult to construct new agent models for each MAS scenario, where agents must have individual objectives and preferences and make sound decisions based on dynamically changing conditions. Sufficient data to capture behavioral diversity is not always available, and modeling based on methods like subject interviews is labor-intensive [Hattori 11]. Another challenge is the difficulty of quantitatively representing preferences and explicitly defining the criteria for decision-making. For instance, it is hard to devise a precise numerical conversion for a preference like "a love for numbingly spicy Sichuan food," which complicates the design of a corresponding evaluation function.

This paper proposes an agent modeling method that leverages Large Language Models (LLMs) to build agents that simulate human behavior at a lower development cost. Our approach creates agents with diverse, plausible behaviors and context-aware decision-making. As a form of collective intelligence, LLMs are expected to embody a wide range of information about the diverse behaviors humans exhibit in various real-world situations. They also enable the use of information in its linguistic form, facilitating decision-making that incorporates preferences and criteria that are difficult to quantify. We propose a method to implement and utilize agent models in MAS by connecting them to an LLM, which allows agents to determine their actions via the LLM during the simulation. Specifically, we implement a MAS where pedestrian-like agents populate a virtual restaurant district and each agent plans and executes actions. By integrating these agents with an LLM, we enable them to determine preferred actions through a natural language-based thought process. We then examine how the behavior of these agents varies based on their individually assigned attributes.

2 章目の翻訳

2. Related Work

2.1. MAS and Agent Modeling for Human Mobility

Multi-Agent Simulation (MAS) has been extensively researched for simulating human mobility in environments like cities, with traffic simulation being a prominent application. The ASEP model is a well-known example that represents vehicle movement to analyze phenomena such as traffic congestion [Nagel 92]. In this model, roads are a series of cells, and vehicles are objects that move discretely between them. The state of the road is updated by moving vehicle-like objects forward according to simple rules, thereby reproducing traffic flow. The ASEP model can approximate actual traffic congestion [Nishinari 06]. However, because it represents the continuous movement of vehicles discretely and processes their interactions based on simple rules, the model is highly abstract and diverges from reality. In contrast, other work observes human driving in realistic environments using driving simulators to develop computational models that capture human decision-making in specific contexts [Hattori 10, 11]. Although this approach effectively reproduces human driving behavior, its high modeling cost—due to setting up the simulator, conducting numerous subject interviews, and analyzing the data—makes it difficult to apply to every MAS target. Other studies focus specifically on taxis, modeling their movements based on accumulated probe data [Cheng 11, Kumar 23]. However, this method is limited to assigning parameters based on statistical data from aggregated driving records and fails to represent the individual styles of each driver. The resulting agent models exhibit uniform behavior based on rational decision-making.

[Fujii 11] incorporates pedestrians into vehicle traffic to conduct MAS of mixed traffic based on their interactions. This work proposes an extension of the Social Force Model to generate plausible pedestrian flow in mixed-traffic environments. Pedestrian agents continuously calculate their direction and speed during the simulation to produce natural crowd movement. Each agent is assigned only spatial information, such as its position and data for detecting contact with other agents, and lacks individual preferences. While this method does not express the diversity of pedestrian behavior, it enables the efficient execution of mixed-traffic simulations. Similarly, [Yamashita 12], which also models pedestrians, proposes a model that compresses 2D pedestrian movement into 1D information. This highly abstract approach prioritizes computational speed over expressive power.

In contrast to these related works that prioritize development cost or execution efficiency, this paper adopts an approach that aims to represent the diversity of human behavior.

2.2. LLM-based Agent Models

Using Large Language Models (LLMs) like ChatGPT to build agents that simulate realistic human societies has gained significant attention. In [Park 23], Park et al. developed agents that perform human-like behaviors and planning by incorporating a unique memory system that extends an LLM. Specifically, they implemented agent functionalities for a "Memory Stream," which represents the agent's memory; "Reflection," for introspecting on its experiences; and "Planning," for creating future action plans. By implementing these three functions in coordination with an LLM, the agents can perform actions consistent with their accumulated experiences and surrounding environment throughout the simulation. Park et al. successfully placed 25 agents in a virtual village and demonstrated that these agents could construct an artificial society through their experiences and natural language interactions. In [Suzuki 24], researchers developed LLM-based agent models to express personality and used MAS to analyze how personality affects the cooperative behavior of a group.

These studies demonstrate the feasibility of creating agent populations with LLM-based agents that can plausibly simulate human groups, which continuously evolve through ongoing decision-making and interaction. In this paper, we build LLM-integrated agents by extending their architecture to enable them to act based on individual preferences and behavioral history. Our approach is designed with scalability in mind for application to larger-scale MAS.

この英文（LLMに基づくエージェントモデルのマルチエージェントシミュレーションへの導入）を用いた理由

自分の研究でLLMを用いたマルチエージェントシミュレーションの開発を行う可能性があるため。

使用したAI：google AI Studio

理由、複数のAIを用いて論文を翻訳し、それぞれのAIでどの論文が良いかを添削させた結果、google AI Studioが一番好評だった。

どのように日本語論文を翻訳させたか？

授業内で説明のあった、「最小主義」に基づいた。

以下の箇条書きした点をAIに注意させた。

- 曖昧な頻度などの表現は、具体的な数値に置き換え可能な場合はそれに置き換える。
- 動詞中心の表現（ofなどの使いまわしを削減）
- 名刺構文の動詞化→「修正を行う」などは「修正する」など
- 重複表現の削除
- 無駄な言葉や冗長な言い回しの排除
- 受動態よりも能動態を使用する

AIに英訳するときに気を付けた点を聞いてみた結果

1. 動詞中心の表現と能動態の積極的な使用

日本語の受動態や名詞構文（「～が行われる」「～が示されている」）を、英語では主体を明確にした能動態と動詞中心の表現に変換しました。これにより、文章がよりダイレクトで力強い印象になります。

- **原文例**: 「～について分析が行われている」
- **翻訳**: "researchers analyze..." / "studies analyze..."
  - 「行われている」という受動的な表現ではなく、「誰が/何が」分析しているのかを明確にし、能動態で表現しました。
- **原文例**: 「～の実現可能性が示されてきている」
- **翻訳**: "These studies demonstrate the feasibility of..."
  - 「示されている」の主語である「これまでの研究」を明確にし、"demonstrate" という力強い動詞を用いています。

2. 冗長な名詞構文の簡略化

日本語で多用される「～のモデル化」「～を行う」といった名詞構文を、シンプルな動詞に置き換えました。これは「最小主義の徹底」にも繋がります。

- **原文例**: 「現実的な運転者の計算モデル化を図っている」

- **翻訳**: "to develop computational models of realistic drivers"
    - 「モデル化を図る」を "develop models" と簡潔に表現しました。
- **原文例**: 「〜アプローチを採る」
- **翻訳**: "this paper adopts an approach that..."
    - より直接的な表現にすることで、文意を明確にしました。

3. 日本語特有の冗長な言い回しの排除

「〜と言える」「〜に成功している」といった、実質的な情報量が少ない、あるいは回りくどい表現を排除し、より簡潔で断定的な表現にしました。

- **原文例**: 「現実からの乖離があると言える」
- **翻訳**: "the model is highly abstract and diverges from reality"
    - 「〜と言える」を削り、「乖離がある」という事実を "diverges from" と直接的に記述しました。
- **原文例**: 「人工的な社会を構築することに成功している」
- **翻訳**: "successfully placed 25 agents... and demonstrated that these agents could construct an artificial society"
    - 「成功した」という結果よりも、具体的に何をしたのか（配置し、実証した）を記述することで、客観性と具体性を高めました。

4. 学術・技術文脈に沿った単語選択

専門用語はもちろんのこと、文脈に合ったフォーマルで的確な単語を選びました。

- **原文例**: 「尤もらしい」
- **翻訳**: "plausible"
    - 学術的な文脈で「もっともらしい」「説得力のある」という意味で頻繁に使われる単語です。
- **原文例**: 「乖離」
- **翻訳**: "divergence" / "diverges from"
    - モデルと現実の「ずれ」を表すのに的確な技術的表現です。
- **原文例**: 「隘路（あいろ）」
- **翻訳**: "A key challenge"
    - 直訳の "bottleneck" も可能ですが、文脈上「モデリングにおける根本的な課題」を指しているため、"A key challenge" の方がより広く意味を捉え、自然な表現となります。

これらの点を総合的に考慮し、原文の意図を正確に伝えつつ、技術文書として明確・簡潔・フォーマルな英文となるよう翻訳を作成しました。